

Course introduction

Outline:

- 1) Syllabus
- 2) Deductive vs inductive reasoning
- 3) The problem of induction
- 4) Coin flipping

Deductive vs inductive reasoning

Deduction:

Drawing inferences that follow logically from premises

Ex: (1) All real, symmetric matrices have real eigenvalues
(2) A is a symmetric matrix
Therefore, A has real eigenvalues

Ex: (1) No one in my daughter's preschool class has a peanut allergy
(2) Zoe is in my daughter's preschool class
Therefore, Zoe is not allergic to peanuts

Risk-free:

Valid arguments + true premises $\rightarrow$ true conclusions
(Conclusion could be wrong if a premise is wrong)

Deductive reasoning can involve probability:

(1) This die has six faces: 1, 2, ..., 6

(2) Each side is equally likely

Therefore, the chance of rolling 4 is $\frac{1}{6}$

Induction: Observations $\rightarrow$ general claims

Risky! We can (and will sometimes) be wrong

Ex. (1) I ate a free sample strawberry at the supermarket
(2) It was ripe and delicious.

Therefore, I should (probably) buy a whole carton.

(Would be more convincing with a random sample)

Ex: (1) Water at 1 atm pressure has always been observed to boil at 100°C

Therefore, all water at 1 atm (probably) boils at 100°C

Ex: (1) I flipped this coin 1000 times and got 502 heads

Therefore, it (probably) has about a 50% chance of landing heads

Statistics: mathematical science of inductive reasoning

The Problem of Induction

Inductive reasoning is not logically valid

Hume: All inductive reasoning requires a presumption that unobserved cases will be like observed cases

How can we justify this?

Logical proof? There is none!*

Past observation? Circular!

Hume admitted we must reason inductively all the time
Called it "custom" or "habit"

Statistics evades this problem in one of two ways

Idea 1: Bayesian reasoning

Whatever our prior beliefs, we know how to update them in light of experience.

Idea 2: "Inductive behavior" (frequentist statistics)

If observations are from a reasonable experiment, we can design methods that give correct conclusions with high probability (provably!)

Coin Flipping

Diaconis, Holmes, & Montgomery (2007):

A coin is a bit more likely to land on the side it started on

Based on physical model (precession)

Predicted $\sim 51\%$ for typical human flipper

Bartoš et al. (2023):

350,757 coin flips

48 flippers using coins from 46 countries

Found 178,079 same-side outcomes ($\approx 50.8\%$)

Frequentist analysis

95% confidence interval $[50.6\%, 50.9\%]$

Based on binomial model:

- Every flip has same probability θ
- Flips are independent

$$\Rightarrow \mathbb{P}_{\theta}(X \text{ same side flips}) = \frac{n!}{x!(n-x)!} x^{\theta} (n-x)^{1-\theta}$$

Confidence interval construction mathematically ensures

$$\mathbb{P}_{\theta}(CI(X) \text{ covers } \theta) \geq 95\% \quad (\text{here, almost } =)$$

Bayesian analysis

95% credible interval [50.6%, 50.9%]

Same binomial model, plus $\text{Unif}[0, 1]$ prior on θ

(Question: whose prior opinion was that?)

Binomial model was wrong!

- Different flippers had different probabilities
- Most flippers improved (got closer to 50%) over time

Kinds of questions we'll ask in this course

Bayesian and frequentist frameworks

What are pros & cons of each?

Where does the prior come from? (Does it matter?)

What does probability mean in each framework?

Sufficiency:

Both analyses summarized data as " $X = 178,079$ "

Did we lose anything? (Not under binomial model)

What about the model structure lets us do this?

Estimation: What's a good way to estimate:

- 1) Overall θ in binomial model
no single best estimator for all θ
 X/n "obvious choice" if $n = 350k$ (unless $\theta = 10^{-6}$)
less obvious for $n = 35$
- 2) θ_i for individual flipper i
new model: $X_i \stackrel{\text{ind.}}{\sim} \text{Binom}(n_i, \theta_i)$ for $i = 1, \dots, m$
how to use data from other flippers?
- 3) How fast $\theta_{i,t}$ changes from flip 1 to flip n_i
variety of possible models
parametric & nonparametric

Testing: How do we efficiently test hypotheses like

- 1) $H_0: \theta \leq 50\%$ vs $H_1: \theta > 50\%$ (one-sided)
Unique best test exists
- 2) $H_0: \theta = 50\%$ vs. $H_1: \theta \neq 50\%$ (two-sided)
Natural choice exists
- 3) $H_0: \theta_i = \theta$ for all flippers vs. $H_1: \theta_i$ varies
Nuisance parameter θ affects null distribution
of any test statistic
Many ways for $(\theta_1, \dots, \theta_m)$ to be non-null
should affect choice of test!
- 4) $H_0: \theta_i$ constant through time for all flippers
vs $H_1: \theta_{i,t}$ tends to be decreasing in t
(Can test this nonparametrically)

Asymptotics : No one calculated 350,757!

Actual model: $X \sim N(n\theta, n\theta(1-\theta))$

(or $X_i \overset{\text{ind}}{\sim} N(n_i\theta_i, n_i\theta_i(1-\theta_i))$)

Want good asymptotic approximations to other models